

Deep reinforcement learning-based beam hopping algorithm in multibeam satellite systems

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Abstract: Beam hopping (BH) is the key technology to improve the system throughput and decrease the transmission delay in multibeam satellite systems. The objective of this study is to find a policy to maximise the expected long-term resource utilisation. The BH illumination plan (BHIP) optimisation problem aimed at minimising the transmission delay is formulated and modelled as a partially observable Markov decision process. To tackle the issue of unknown dynamics and prohibitive computation, an artificial intelligence method named deep reinforcement learning (DRL) is first proposed to solve the BHIP problem in multibeam satellite systems. The proposed DRL-BHIP algorithm considers a series of realistic conditions, including the traffic demands in spatial distribution and temporal variation, ModCod constraints, antenna radiation pattern and inter-beam interference. The state reformulation concept is adopted to characterise the traffic spatial and temporal features. Simulation results show that the proposed DRL-BHIP algorithm can decrease the transmission delay and improve the system throughput compared with existing algorithms.

1 Introduction

The multibeam satellite systems have been making efforts on improving the system performance in order to follow the trends of terrestrial networks in terms of peak data rates and service costs. Through a large number of beams and flexible on-board payloads, the multibeam satellites, especially for the high-throughput satellites have improved the system performance effectively and attracted considerable interests. Considering the non-uniform traffic demands among beams, the Geostationary Earth Orbit (GEO) multibeam satellites need dynamic resource allocation (DRA) technique to match the offered traffic with the required traffic among the beams, so that the traffic can be transmitted in real-time [1–3].

The DRA in GEO multibeam satellite systems can be classified into the flexible resource allocation (FRA) and beam hopping (BH). For FRA, the power allocation [4–7] and bandwidth allocation [8–11] are studied. Choi and Chan *et al.* optimised the power allocation in order to adapt to channel conditions and meet traffic demands through Lagrange multiplier method [4]. However, co-channel interference (CCI), which constitutes a major limitation in multibeam satellite systems, is not considered. The consideration of CCI in [5] gives rise to a numerical approach for the interference mitigation problem, since the closed form of the relevant solution is hard to obtain. Further study is taken in [6] which proves the NP-Hardness of power allocation with CCI and decomposes the problem into independent colour-based sub-problems rendering the proposed two-stage Genetic Algorithm (GA) and Simulated Annealing metaheuristics algorithm. Similar to the work in [6], a metaheuristics algorithm named particle swarm optimisation based DRA is proposed in [7]. The work in [8] dealt with the problem of quality of service (QoS) provisioning for packet traffic by considering the bandwidth allocation techniques. Lei and Vazquez-Castro [9] proposed a carrier allocation algorithm based on the binary power allocation to reach the signal to interference plus noise (SINR) balancing. Focusing on the integration of hybrid GEO and low earth orbit satellites, bandwidth resource allocation strategy [10] and cooperative transmission [11] for GEO relay systems are proposed.

With the development of GEO multibeam satellite systems, the beamwidth is smaller and the number of beams is larger in the order of hundreds to thousands. Large number of beams would increase the payload mass of the satellites and the DC power consumption. In this context, the BH technique emerged, where the multibeam satellite dynamically illuminates the cells with a small number of active beams. Many scholars have conducted the BH in satellite systems [12–23]. Angeletti *et al.* [12] assessed the advantages in terms of system performance brought by the utilisation of BH technique and proposed GA based BH illumination plan (BHIP) optimisation. Anzalchi *et al.* [13] further investigated the advantages of BH and compared the performance with non-hopped systems. To achieve the next generation of Terabit/s satellites, the use of BH to reduce the number of transponder amplifiers required is addressed [14]. To decrease the computation complexity of the algorithms in [12, 13], two heuristic algorithms, which allocate capacity resources based on the per-beam traffic requests, are designed [15]. Considering that the algorithms in [12–15] only focused on the long-term traffic variations, Alegret *et al.* [16] proposed two mechanisms to deal with both the long-term and short-term traffic variations. Liu *et al.* [17] studied a new method of rate control on the incoming traffic in the BH multibeam satellite systems. Different from works in [12–17] where the focus is on the capacity optimisation, Han *et al.* [18] considered the delay fairness for the QoS equilibrium in BH satellites. To further improve the efficiency of on-board resource utilisation, joint BH and FRA [19], joint BH and precoding [20] schemes are proposed to improve the system performance.

Although the existing DRA algorithms made great achievements in throughput improvement and delay reduction, the GA or other heuristic algorithms need to be revised once the system scenarios change slightly. In other words, these off-line algorithms result in high computational complexity which is not appropriate for on-board resource management. As an online optimisation method, the deep reinforcement learning (DRL) has broad applications in sequential decision-making problems [21–24] and has been adopted in solving many flexible resource scheduling problems [25–27]. The key idea of the DRL is to make better decisions by learning from the historical high-dimensional environments. The problem of beam allocation in the BHIP is a

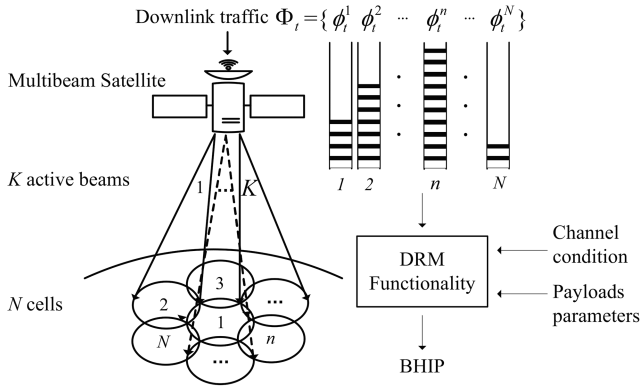


Fig. 1 Illustration of BH multibeam satellite systems

dynamic resource optimisation problem. For BHIP problems, the beam allocation decisions depend not only on the traffic loads and the channel condition of each beam but also on the interrelationship among illuminated beams. Meanwhile, the BHIP decision should guarantee that the total transmitting power does not exceed the maximum satellite power. Last but not least, the former illumination decisions have impacts on the latter decisions, which makes the BHIP belong to the sequential decision-making problems. To this end, a DRL based BHIP (DRL-BHIP) is studied and proposed.

In this paper, the optimisation problem, which aims at minimising the transmission delay, is formulated. The formulated optimisation problem is further modelled as the Markov decision process (MDP), where a novel state reformulation technique is adapted to appropriately represent the time-varying states. To maximise the long-term accumulated rewards of the modelled MDP, an artificial intelligence method named DRL is first proposed to solve the BHIP problem in multibeam satellite systems. To tackle the issue of instability and divergence of the DRL-BHIP, the target network and replay memory techniques are adopted. At last, the computer simulations are conducted to validate the proposed DRL-BHIP algorithm in terms of transmission delay, by comparing with existing algorithms.

Main contributions of this paper are summarised as follows:

- The DRL method is first adapted to solve the BHIP problem to minimise the transmission delay, where the formulated optimisation problem is further modelled as the MDP.
- The state reformulation concept is adopted where the traffic information among beams is reformulated into a multi-dimensional tensor so that the spatial-temporal feature of traffic can be characterised. Meanwhile, the reward is designed based on the delay-aware matrix to minimise the expected long-term transmission delay.
- Simulation results show the effectiveness of the proposed DRL-BHIP algorithm in terms of delay reduction and system throughput, where the realistic conditions are considered, including the payloads parameters, traffic demands in spatial distribution and temporal variation, ModCod constraints and inter-beam interference.

The rest of the paper is organised as follows: Section 2 introduces the system model of the multibeam satellite systems with BH and formulates the optimisation problem. Section 3 describes the proposed DRL-BHIP algorithm. Simulation result and analysis are given in Section 4. Section 5 concludes the paper.

2 System model and problem formulation

In this section, we present the system model on multibeam satellite systems with BH. Then, the BHIP optimisation problem is formulated.

2.1 System model

This paper considers the forward link of multibeam transparent satellite systems for providing broadband Internet access services to user terminals (UTs). The UTs are connected transparently through the MBS to gateway stations. The focus of this paper is on the satellite-to-user link (downlink), where the BH concept is applied. Without loss of generality, K active beams are supposed to illuminate the N cells covering the area, where $K \leq N$. Denote the K active beams as $\mathcal{B} = \{b_k | k = 1, 2, \dots, K\}$, and the N cells as $\mathcal{C} = \{c_n | n = 1, 2, \dots, N\}$. The considered multibeam satellite system is shown in Fig. 1.

The time slot to dynamically illuminate N cells is denoted as t_s , and the illuminated cells are assigned the entire bandwidth W . The BHIP at time slot t_j can be denoted as $X_{t_j} = [x_{t_j}^{c_1}, x_{t_j}^{c_2}, \dots, x_{t_j}^{c_N}]^T$, where the $x_{t_j}^{c_n} = 1$ represents the cell c_n is illuminated at time slot t_j and the $x_{t_j}^{c_n} = 0$ otherwise. As shown in Fig. 1, the BHIP is implemented through the dynamic resource management functionality, where the decision is made based on the packets in satellite buffer, the channel condition and payloads parameters.

For an illuminated cell c_n , that is $x_{t_j}^{c_n} = 1$, the SINR of cell c_n at time slot t_j can be expressed as

$$\Gamma_{t_j}^{c_n} = \frac{h_{t_j}^{c_n c_n} \cdot P_{t_j}^{c_n}}{N_0 \cdot W + \sum_{c_i \in \mathcal{C}, c_i \neq c_n, x_{t_j}^{c_i} = 1} h_{t_j}^{c_n c_i} \cdot P_{t_j}^{c_i}} \quad (1)$$

where $h_{t_j}^{c_n c_i}$ is the power gain of intended cell c_i from the transmitting beam c_i at time slot t_j , which mainly includes the transmitting antenna gain, the free space loss and the receiving antenna gain. $P_{t_j}^{c_n}$ is the transmitting power to cell c_n at time slot t_j . N_0 is the noise power spectral density which depends on the noise temperature of the receiver antenna [26].

Based on the SINR calculated through (1), the offered capacity for cell c_n can be expressed as

$$C_{t_j}^{c_n} = x_{t_j}^{c_n} \cdot W \cdot f_{\text{DVB-S2}}(\Gamma_{t_j}^{c_n}) \quad (2)$$

where the $f_{\text{DVB-S2}}$ is the performance mapping function based on the digital video broadcasting through satellite second generation (DVB-S2) specifications [28].

Traffic among cells is scheduled by optimising the BHIP so that the transmission delay can be minimised. Here, the transmission delay is defined as the average packet delay, which is the time interval between packet arrival time and departure time. Assuming the packet arrival follows Poisson process with arrival rates $\Lambda_{t_j} = \{\lambda_{t_j}^{c_n} | c_n \in \mathcal{C}\}$, where the $\lambda_{t_j}^{c_n}$ is the packets arrival rate for cell c_n during the time period $[t_j, t_{j+1}]$. For a given packet p_i , the packet arrival time is denoted as $t_{p_i}^a$. Without loss of generality, the size of packets is assumed to be equal which can be denoted as M bits. Denote the packets in the satellite buffer at time t_j as $\Phi_{t_j} = \{\phi_{t_j}^{c_n} | c_n \in \mathcal{C}\}$, where the $\phi_{t_j}^{c_n}$ is the set of packets in buffer to be transmitted for cell c_n at time t_j . Denote the maximum delay of packets as T_{th} , that is to say, the packets with delay more than T_{th} will be dropped.

Then, the packets set Φ_{t_j} in buffer is determined by the packets set of previous time slot $\Phi_{t_{j-1}}$, the BHIP $X_{t_{j-1}}$, the channel condition $H_{t_{j-1}}$ and the packets arrival rates $\Lambda_{t_{j-1}}$. For simplicity, the packets in each cell are transmitted in the first-come-first-serve (FCFS) mode. Thus, the set of packets at time slot t_j can be expressed as

$$\Phi_{t_j} = g(\Phi_{t_{j-1}}, X_{t_{j-1}}, H_{t_{j-1}}, \Lambda_{t_{j-1}}) \quad (3)$$

where the function $g(\cdot)$ is the packet scheduling process with FCFS.

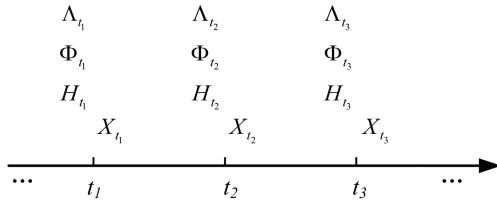


Fig. 2 Illustration of BH illumination problem (BHIP)

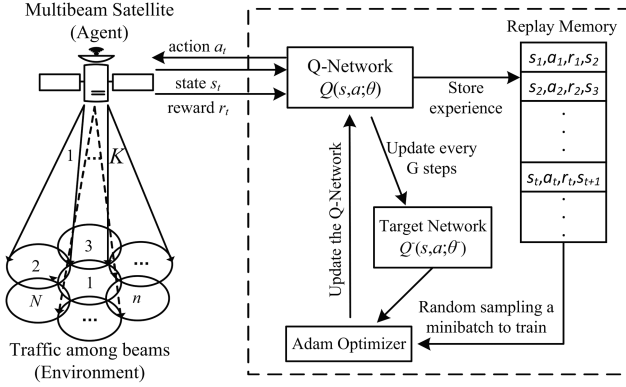


Fig. 3 Architecture of the proposed DRL-BHIP algorithm

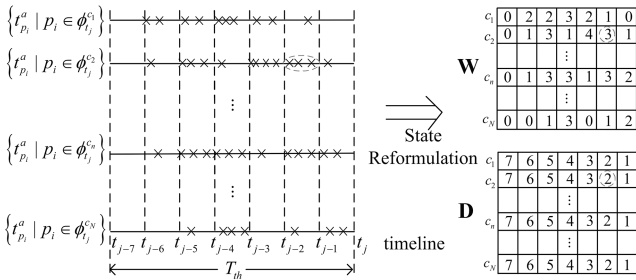


Fig. 4 Illustration of state reformulation process

The packet scheduling process with BH in multibeam satellite systems is illustrated in Fig. 2.

2.2 Problem formulation

In this paper, the aim is to minimise the transmission delay by optimising the BHIP scheme. More specifically, the objective function is to minimise the transmission delay while the decision variable is the BHIP schemes X_{t_j} . What's more, the decision should be made based on system current states, namely the packets in buffer Φ_{t_j} , the channel condition H_{t_j} and the packet arrival rate Λ_{t_j} . To this end, the optimisation problem can be formulated as (4) subject to (5)–(8):

$$\text{opt. } \mathcal{P} = \min_{X_{t_j}} \sum_{t_j \in \mathcal{T}} \sum_{c_n \in \mathcal{C}} \sum_{p_i \in \Phi_{t_j}^{c_n}} (t_j - t_{p_i}^a) \quad (4)$$

$$\text{s.t. } \sum_{c_n \in \mathcal{C}} x_{t_j}^{c_n} \leq K, \quad x_{t_j}^{c_n} \in \{0, 1\}, \quad \forall c_n \in \mathcal{C}, t_j \in \mathcal{T} \quad (5)$$

$$\sum_{c_n \in \mathcal{C}} P_{t_j}^{c_n} \leq P_{\text{tot}}, \quad \forall t_j \in \mathcal{T} \quad (6)$$

$$P_{t_j}^{c_n} \leq P_b, \quad \forall c_n \in \mathcal{C}, t_j \in \mathcal{T} \quad (7)$$

$$t_j - t_{p_i}^a \leq T_{\text{th}}, \quad \forall t_j \in \mathcal{T}, p_i \in \Phi_{t_j} \quad (8)$$

where the set $\mathcal{T} = \{t_1, t_2, \dots\}$ is the set of time slots during a period of time.

For the formulated optimisation problem $\mathcal{P}$, constraint in (5) means the number of simultaneously illuminated cells should not

exceed the number of active beams K . The constraint in (6) and (7) represents the transmitting power of all beams or any single beam should not exceed the maximum on-board power and the beam power limitation, respectively. The (8) means that the delay of packets in buffer would not exceed the delay threshold T_{th} . Thus, we know the optimisation problem $\mathcal{P}$ is a sequential decision-making problem with multiple complex constraints. The proposed DRL-BHIP algorithm will be described in the next section.

3 The DRL-BHIP algorithm

The architecture of the DRL-BHIP is illustrated first. Then the detailed designation including the MDP model, Q-Network training, and the action selection are described. Finally, the description of the DRL-BHIP algorithm implementation is given.

3.1 DRL-BHIP architecture

The architecture of the DRL-BHIP algorithm is illustrated in Fig. 3, where the satellite is modelled as the agent while the traffic among beams is modelled as the environment.

From Fig. 3, the MDP is modelled according to the target optimisation problem $\mathcal{P}$. The Q-Network is the action-value function which is responsible for mapping the system state into action values (namely the Q values). Replay memory, target network, and Adam optimiser are used to train the Q-Network to improve the decision-making performance.

3.2 MDP model

As a prominent analytical framework for modelling the sequential decision-making problems, the MDP is a discrete time stochastic control process. More precisely, the MDP consists of a set of states $s \in \mathcal{S}$, a set of actions $a \in \mathcal{A}$, a reward function $r \in \mathbb{R}$, and a series of transition probabilities $p(s_{t+1}|s_t, a_t)$. The goal is to choose a policy so as to maximise some cumulative function of the random rewards, typically the expected discounted sum over a potentially horizon, i.e. $R_t = \sum_{i=0}^{\infty} \gamma^i \cdot r_{t+i}$. It should be noted that the r_{t+i} means the instant reward for time $t+i$ and the $\gamma \in [0, 1]$ represents the discount of expected rewards.

From Fig. 2, we know the BHIP optimisation problem has the Markov proper and it can be modelled as MDP. Considering the packet arrival rates Λ_{t_j} and channel condition H_{t_j} are time-varying and hard to know. Thus, we adopt the deep Q network, which is a model-free DRL method, to improve the decision-making performance with no prior-knowledge on $p(s_{j+1}|s_j, a_j)$.

3.2.1 State. For the optimisation problem $\mathcal{P}$ as (4), the BHIP decision X depends on the current packets in buffer Φ , the channel condition H . Considering the state is the input of the Q-Network which is a deep neural network, the state should be defined reasonably and regularly structured. Here, the state reformulation concept is used in designing the state [25]. The state reformulation process is as follows and illustrated in Fig. 4.

As described in (8), the delay of packets in the buffer Φ_{t_j} is smaller than the delay threshold T_{th} . Then, the arrival time of packets in Φ_{t_j} are located in the time interval $T_{\Phi} = [t_j - T_{\text{th}}, t_j]$. Without loss of generality, the time interval T_{Φ} can be divided into F parts, where f_{th} part means the maximum packets delay is $(F - f + 1) \cdot (T_{\text{th}}/F)$. The f_{th} time interval is calculated as

$$\left[(f + F - 1) \cdot \frac{t_j - T_{\text{th}}}{F}, (f + F) \cdot \frac{t_j - T_{\text{th}}}{F} \right).$$

Then, the number of packets $w_{t_j}^{n,f}$, whose arrival time is located in the f_{th} time interval and to be transmitted for cell n , can be expressed as

$$w_{t_j}^{n,f} = \left| \left\{ p_i \mid p_i \in \Phi_{t_j}^{c_n}, f + F - 1 \leq \frac{t_{p_i}^a - F}{t_j - T_{\text{th}}} \leq f + F \right\} \right| \quad (9)$$

Table 1 The DRL-BHIP Implementation

1. Initialisation
2. Initialise BH multibeam satellite scenarios
3. Initialise the delay and throughput parameters
4. Initialise the experience pool $\mathcal{D}$ to capacity N_{ep}
5. Initialise the Q-Network with random weights θ
6. Initialise the target network with weights $\theta^- = \theta$
7. Training and operating
8. For time slot $t_j \in \mathcal{T}$ do
9. Observe the traffic and abstract the state
10. Reformulate the state into the two-dimensional tensor s_j
11. Set the reward r_j based on the tensor s_j
12. Store the experience tuples $(s_{j-1}, a_{j-1}, r_j, s_j)$ in $\mathcal{D}$
13. If $\text{rand} \leq \epsilon$
14. randomly select K actions $a_j \in \mathcal{A}$
15. Else
16. select K actions with maximum Q values
17. End If
18. Illuminate the selected cells based on the K actions a_j
19. Sample a minibatch of $(s_j, a_j, r_{j+1}, s_{j+1})$ from $\mathcal{D}$
20. Calculate the label Q-value y_j as (14)
21. Calculate the CNN loss $L(\theta)$ as (15)
22. Train the Q-Network in Adam Optimiser
23. Decrease the exploration probability ϵ
24. If $\text{mod}(j, G) = 0$
25. Set the target network $\hat{Q}$ with weights $\theta^- = \theta$
26. End If
27. End For

where the operator $|\cdot|$ is the cardinality of a given set.

In Fig. 4, the marker 'x' represents the packet arrival time. And the packet delay threshold is $T_{th} = 7 \cdot T_s$ and $F = 7$. For example, there are three packets whose arrival instant are located in $[t_{j-2}, t_{j-1}]$ and to be transmitted to cell c_2 . Thus, the delay-aware value $w_{t_j}^{2,6}$ in $\mathbf{W}_{t_j}$ is three, and the value $d_{t_j}^{2,6}$ in $\mathbf{D}_{t_j}$ is two.

To this end, the state in modelled MDP is defined as

$$s_{t_j} = (\mathbf{W}_{t_j}, \mathbf{D}_{t_j}) \quad (10)$$

3.2.2 Action. For the BHIP optimisation problem $\mathcal{P}$ as (4), the decision variable is the BHIP vector X_{t_j} and the aim is to minimise the cumulative delay. Meanwhile, the decision variable should satisfy the constraints (5)–(8), which constitutes the action feasible region. Denote the action feasible region as $\mathcal{A}$, which can be expressed as (11)

$$\mathcal{A} = \{a_1, \dots, a_k, \dots, a_K \mid a_k \in \mathcal{C}, x_{t_j}^{a_k} = 1, \forall a_k\} \quad (11)$$

3.2.3 Reward. For the BHIP optimisation problem $\mathcal{P}$ as (4), the objective aims at minimising the packets delay. Thus, the reward value can be designed as a big value when the current cumulative delay is small. While the reward value will be designed a small value when the current delay is big. Thus, the reward can be designed as

$$r_t = - \|\mathbf{W}_{t_j} \odot \mathbf{D}_{t_j}\| \quad (12)$$

where the operator $\odot$ means the Hadamard product, and the operator $\|\cdot\|$ means the sum of all elements in a matrix.

3.3 DRL-BHIP designation

This section mainly describes the DRL-BHIP designation in terms of the Q-Network structure and training.

3.3.1 Q-Network structure. Q-Network is the action-value function which evaluates the expected accumulative rewards for taking an action under the given states. Different from traditional reinforcement learning such as Q-learning, the action-value function is implemented through a neural network other than the Q-tables. Here, we use a deep convolutional neural network (CNN) to approximate the optimal action-value function as (13) [22].

$$Q^*(s, a) = \max_{\pi} \mathbb{E}[r_t + \gamma r_{t+1} + \dots \mid s_t = s, a_t = a, \pi] \quad (13)$$

3.3.2 Q-Network training. As is known, reinforcement learning tends to be unstable or even to diverge when CNN is adopted to approximate the Q value function [28]. To overcome this disadvantage, the target network Q^- and replay memory $\mathcal{R}$ are adopted in the proposed DRL-BHIP algorithm to improve the stability of the Q-Network. During the Q-Network training, the replay memory $\mathcal{R}$ is cleared during the initialisation phase of the DRL-BHIP algorithm. Then, the newly formed experience tuple $e_t = (s_t, a_t, r_{t+1}, s_{t+1})$ is stacked into $\mathcal{R}$ for each training step. And the DRL-BHIP algorithm starts the training in case that the number of stored experiences in $\mathcal{R}$ exceeds the number N_{st} . During each training step, a batch of N_{mb} experience items, here we call it a minibatch, are randomly sampled from replay memory $\mathcal{R}$ and the target value is calculated through the target network Q^- based on the Bellman equation [29]. The target value is calculated as

$$y_t = \begin{cases} r_{t+1}, & \text{if terminated} \\ r_{t+1} + \gamma \cdot \max_{a \in \mathcal{A}} Q^-(s_{t+1}, a; \theta^-), & \text{otherwise} \end{cases} \quad (14)$$

Based on the target value in (13), the loss $L(\theta)$ of current Q-Network at t_{th} iteration can be calculated as

$$L_t(\theta_t) = \mathbb{E}_{(s, a, r, s') \sim U(\mathcal{R})} [y_t - Q(s, a; \theta_t)]^2 \quad (15)$$

where θ_t are the parameters of the Q-Network at iteration t and θ^- are the target network parameters used to compute the target at iteration t . The target network parameters θ^- are only updated with the Q-Network parameters (θ_t) every G steps and are held fixed between individual updates.

In this paper, the Adam optimiser is used to train the Q network by minimising the loss function $L(\theta)$ [30].

3.4 DRL-BH implementation

The proposed DRL-BHIP algorithm consists of two main stages, the initialisation, and training stages. During the initialisation stage, the Q-Network and target network are initialised with random weights, and the replay memory is emptied. In the training stage, the Q-Network is used to make decisions and trained during each decision-making time slots.

The DRL-BHIP algorithm process is illustrated in Table 1.

4 Simulation results and analysis

Simulation parameters are given and then simulation results are analysed.

4.1 Simulation parameters

For simulation parameters, scenario-related and DRL-related parameters are given.

4.1.1 Scenario parameters. A typical Ka-band multibeam satellite system is assumed, where the operating frequency in downlink is 20 GHz. The total bandwidth of 500 MHz and DVB-S2 specification are used. Multibeam antenna radiation pattern refers to the rec. ITU-S.672 [31], where $\theta_b = 0.28^\circ$, $a = 2.88$, $b = 6.32$ and $L_s = -25$. The antenna radiation pattern is showed as (16) and plotted as Fig. 5.

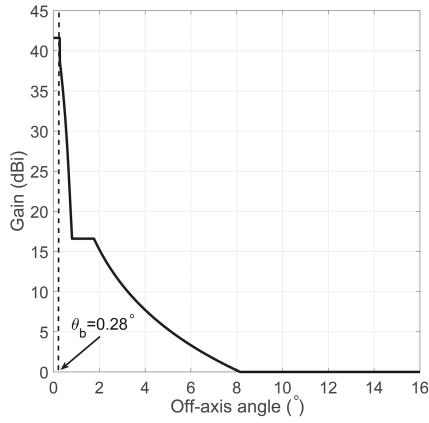


Fig. 5 Multibeam satellite antenna radiation pattern

Table 2 Scenario parameters

Parameters	Values
scenario parameters	
Ka band f_c	20 GHz
system bandwidth B_{tot}	500 MHz
time slot duration t_s	100 ms
number of beams K	10
number of cells N	37
multibeam antenna beamwidth	$\theta_b = 0.56^\circ$
maximum antenna gain G_m	41.6 dBi
total satellite power P_{tot}	33 dBW
maximum beam power P_b	30 dBW
propagation loss	209.5 dB
UT receiving G/T	18.5 dB/K
traffic distribution	illustrated as Fig. 6
traffic time variation	illustrated as Fig. 7
packet size M	100 kbit
packet delay threshold T_{th}	4 s
delay-aware matrix F	40

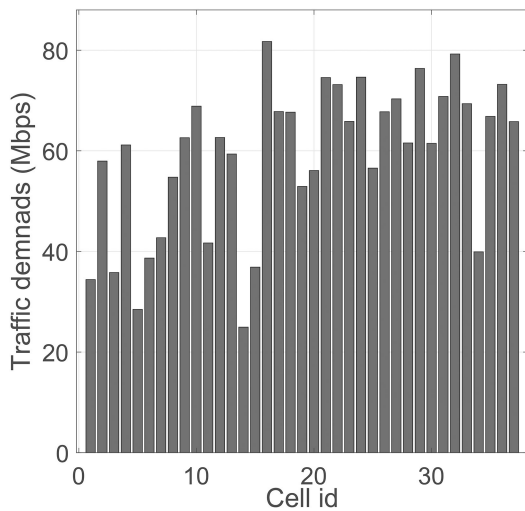


Fig. 6 Traffic demands among each cell

$$G(\theta) = \begin{cases} G_m, & \text{for } \theta < \theta_b \\ G_m - 3(\theta/\theta_b)^2, & \text{for } \theta_b \leq \theta < a\theta_b \\ G_m + L_s, & \text{for } a\theta_b \leq \theta < b\theta_b \\ \max \{G_m + L_s + 20 - 25 \lg(\frac{\theta}{\theta_b}), 0\}, & \text{else} \end{cases} \quad (16)$$

where the θ is the off-axis angle in degrees.

Main scenario parameters are illustrated as Table 2 [13].

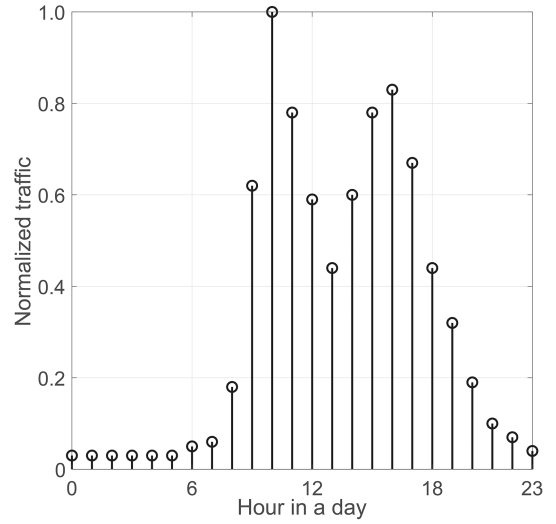


Fig. 7 Traffic temporal variation in a day

Table 3 DRL-BHIP Algorithm parameters

Parameters	Values
algorithm parameters	
training epoches	2000
time slots every epoch	10,000
learning rate	10^{-5}
experience pool	5000
replay start size	1000
minibatch size	16
target network update frequency	100
activation function	Relu
discount factor γ	0.90
initial exploration rate ϵ_i	0.5
final exploration rate ϵ_f	0.01

Traffic model in terms of spatial distribution and temporal variation is illustrated as Figs. 6 and 7 [27]. In Fig. 6, the traffic demands in each cell represents the peak traffic while Fig. 7 shows the assumed profile of traffic variation over 24 h.

4.1.2 DRL-BHIP algorithm parameters.: In the DRL-BHIP algorithm, total 2000 epochs are trained to evaluate the performance of proposed DRL-BHIP algorithm. Other detailed parameters are given in Table 3 [22]. The structure of Q-Network and target network is illustrated as Fig. 8, where the Q-Network consists of two convolutional network layers and three fullyconnected layers.

4.2 Simulation result and analysis

In this section, algorithms performance in terms of transmission delay and system throughput is evaluated. We compared the proposed DRL-BHIP algorithm with the following algorithms:

- *Rand*: where the K active beams are randomly selected to illuminate the coverage.
- *FixTime*: where fixed time slots are allocated to each beam based on the demanding traffics.
- *MaxMin*: where maxmin rate control method for BH proposed in [17].
- *GA*: where GA is adopted to optimise the BHIP scheme.

4.2.1 Transmission delay performance.: In this case, the traffic of each beam follows the non-uniform distribution as illustrated in Fig. 6. We evaluate the performance of transmission delay under different beam traffic loads. The simulation result is illustrated in Fig. 9.

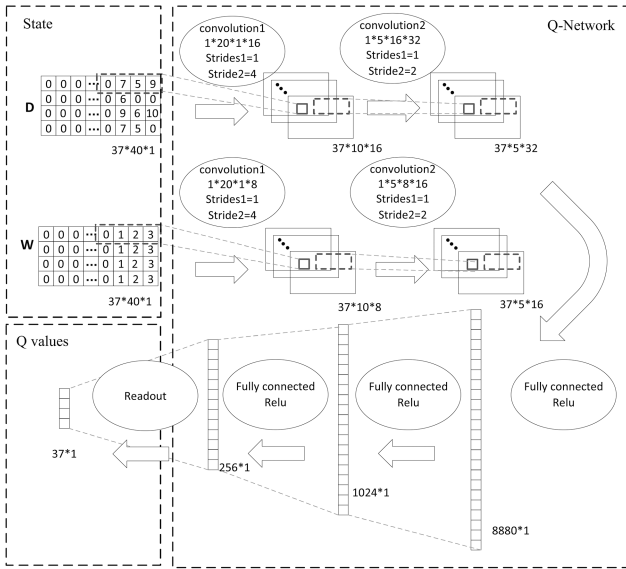


Fig. 8 Structure of the Q-Network

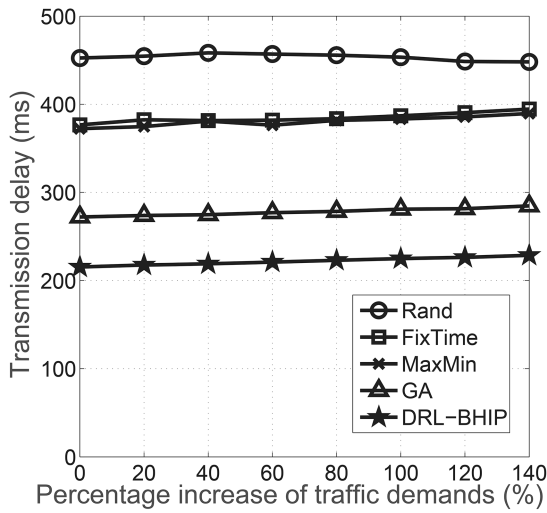


Fig. 9 Transmission delay under different traffic demands

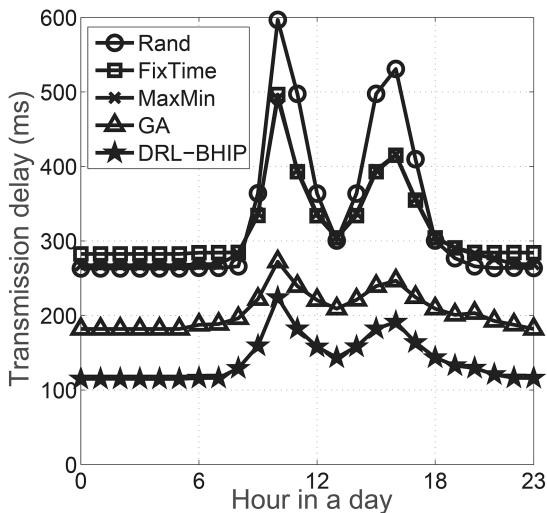


Fig. 10 Transmission delay under time variations in a day

It can be seen from Fig. 9 that the transmission delay of the proposed DRL-BHIP algorithm is smaller than that of the Rand, FixTime, MaxMin, and GA algorithms. For example, under the scenario of 80% traffic increase, the transmission delay of the Rand, FixTime, MaxMin, GA, and DRL-BHIP algorithm is $t = 460$ ms, $t = 380$ ms, and $t = 380$ ms, $t = 280$ ms and

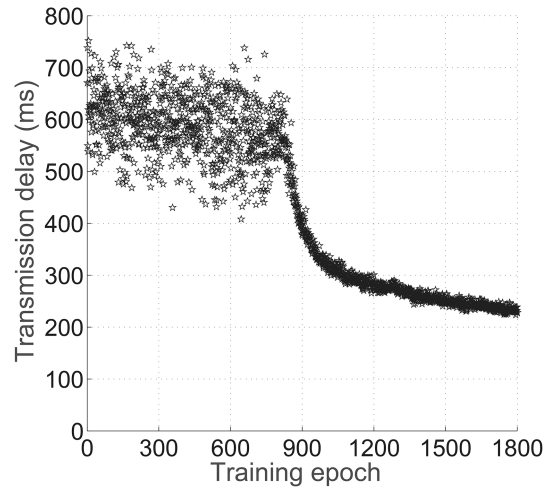


Fig. 11 Transmission delay during training process

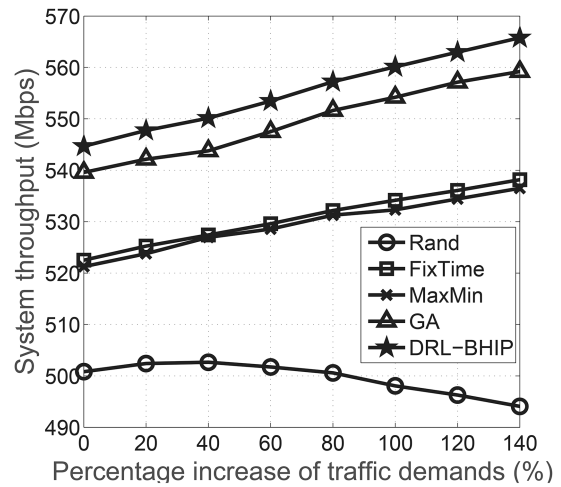


Fig. 12 System throughput under different traffic demands

$t = 220$ ms, respectively. In other words, the proposed DRL-BHIP algorithm can decrease the transmission delay in 52.2, 42.1, 42.1, and 21.4%, compared with the Rand, FixTime, MaxMin, and GA, respectively. Fig. 10 shows that the transmission delay varies with the hours in a day. From Fig. 10, the transmission delay is smaller than the other four algorithms.

The convergence performance of the DRL-BHIP is validated in Fig. 11, where the simulation parameters are the same as that of Fig. 10 at 10 am. From Fig. 11, the proposed DRL-BHIP algorithm converged after about 1500 training periods. We can know that the DRL-BHIP can converge in relatively small training steps making the proposed DRL-BHIP appropriate in time-varying traffic situations.

4.2.2 System throughput performance. We evaluate the performance of system throughput under different beam traffic loads in Fig. 12.

From Fig. 12, it can be seen that the proposed DRL-BHIP algorithm achieves higher system throughput compared with the Rand, FixTime, MaxMin, and GA algorithms. For example, under the scenario of 80% traffic increase, the system throughput of the Rand, FixTime, MaxMin, GA, and DRL-BHIP algorithm is 500, 532, 531, 552, and 557 Mbps, respectively. That is to say, the proposed DRL-BHIP algorithm can improve the system throughput in about 11.4, 4.7, 4.9, and 0.9%, compared with the Rand, FixTime, MaxMin, and GA, respectively. Fig. 13 shows convergence performance in terms of system throughput.

5 Conclusion

In this paper, a DRL-BHIP algorithm is proposed in multibeam satellite systems. The novelty lies in the combination of artificial

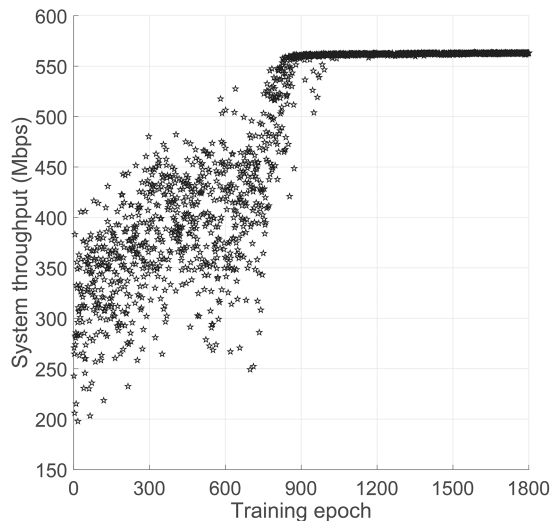


Fig. 13 System throughput during training process

intelligence algorithm and resource management technique, where the state reformulation and reward designation are specified. Comparing with the existing algorithms, the proposed DRL-BHIP algorithm can decrease the transmission delay and improve the system throughput by up to 52.2 and 11.4%, respectively. The proposed DRL-BHIP algorithm sets a general framework allowing the DRA problem into the artificial intelligence methods, making it feasible to practically implement the on-board online DRA approach in multibeam satellite systems.

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7 References

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